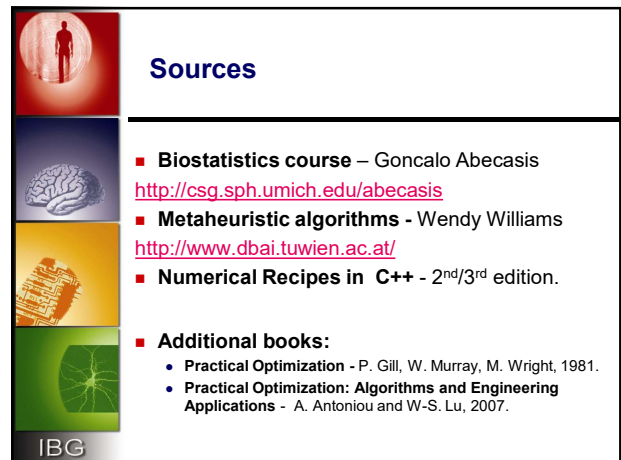



**Signal & Data Analysis in Neuroscience
2020
Optimization**

Izhar Bar-Gad
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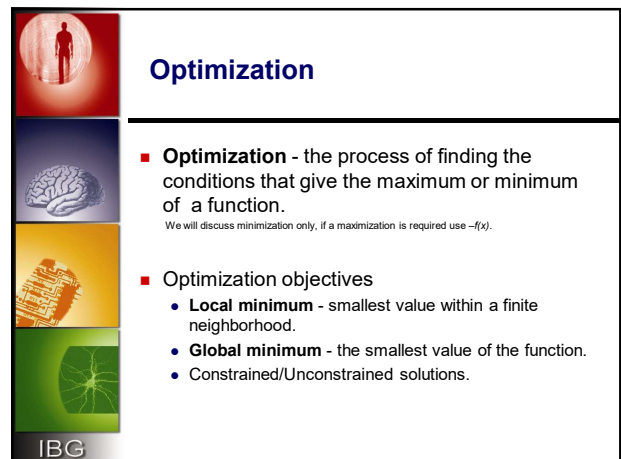


Sources

- **Biostatistics course** – Goncalo Abecasis
<http://csg.sph.umich.edu/abecasis>
- **Metaheuristic algorithms** - Wendy Williams
<http://www.dbai.tuwien.ac.at/>
- **Numerical Recipes in C++** - 2nd/3rd edition.
- **Additional books:**
 - **Practical Optimization** - P. Gill, W. Murray, M. Wright, 1981.
 - **Practical Optimization: Algorithms and Engineering Applications** - A. Antoniou and W-S. Lu, 2007.

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Optimization

- **Optimization** - the process of finding the conditions that give the maximum or minimum of a function.
We will discuss minimization only. If a maximization is required use $-f(x)$.
- **Optimization objectives**
 - **Local minimum** - smallest value within a finite neighborhood.
 - **Global minimum** - the smallest value of the function.
 - Constrained/Unconstrained solutions.

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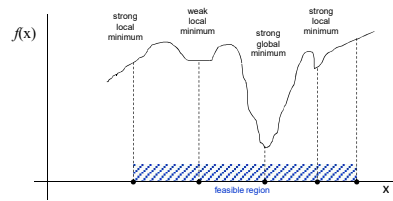
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Types of minima



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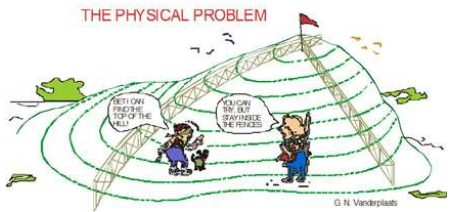



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Constraints

$$\begin{aligned} &\min f(\mathbf{x}) \\ &\text{subject to } g_i(\mathbf{x}) = c_i \quad \text{for } i = 1, \dots, n \quad \text{Equality constraints} \\ &\quad h_j(\mathbf{x}) \geq d_j \quad \text{for } j = 1, \dots, m \quad \text{Inequality constraints} \end{aligned}$$

THE PHYSICAL PROBLEM



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Optimization

- Univariate methods
- Multivariate methods
- Simulated annealing
- Genetic algorithms

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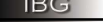



Optimization – single variable

Finite (small) set

- Finite (typically small) set of alternatives
→ scoring methods.
- Example: minimize the cost of a course, choose between two options: SDA or MATLAB.

Course	Effort	Gain	Cost = Effort * 7 - Gain
SDA	2	10	4
MATLAB	1	0	7



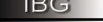





Optimization – single variable

Infinite set

- Infinite (or finite but very large) →
Analytic and numeric optimization methods
- Example: Find the minimal value of $f(x) = x^2 + 3x + 4$ under the constraint $g(x) = x^3 > 4$



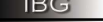





Optimization – single variable

Analytic solution

- Calculate the derivative. Identify all the zero values and calculate the second derivative to identify local minima.
- Compare the values of all the local minima to identify the global minima.
- Example:
 $f(x) = Ax^2 + Bx + C \rightarrow f'(x) = 2Ax + b$
Extremum: $2Ax + B = 0 \rightarrow x = -B/2A$
Minimum: $A > 0 \rightarrow f''(x) = 2A > 0$



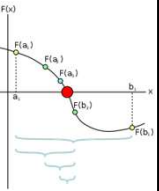





Detour – Numeric solutions

Zero crossing

- Assume that you know:
 - Point a where $f(a)$ is positive
 - Point b where $f(b)$ is negative
 - $f(x)$ is continuous between a and b
- Find x such that $f(x)=0$
(actually within $[-\epsilon, +\epsilon]$)
- Simple segmentation algorithm:
bisection - divide the interval the interval during each iteration.



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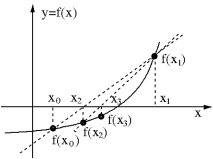



Detour – Numeric solutions

Segmenting via approximation (interpolation)

$$f^*(x) = f(a) + (x - a) \cdot \frac{f(b) - f(a)}{b - a}$$

- Select new points using $f^*(x)=0$
- Finding the zero of $\sin(x)$ in the interval $[\pi/4, \pi/2]$ with accuracy of 10^{-5} requires 17 calls using the bisection method but only 5 using the interpolation method.



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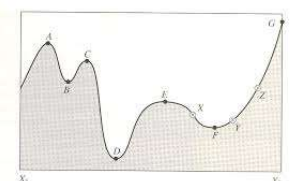





Optimization - Numeric solution

Bracketing

- Bracket minimum
 - Find 3 points such that
 - $a < b < c$
 - $f(b) < f(a)$ and $f(b) < f(c)$
- Tighten bracketing interval



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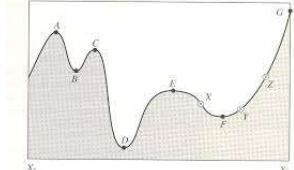





Optimization - Numeric solution

Initial bracketing interval

- Find two point: $a, b \rightarrow f(a) > f(b)$
- Take successively larger steps beyond b until function increases.



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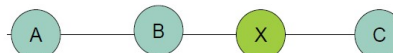



Bracketing

Reducing the interval



- What is the best location for a new point X ?



- We want to minimize the size of the next search interval: either from A to X (for $f(X) > f(B)$) or from B to C (for $f(X) < f(B)$)

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Bracketing

Golden section

$$w = \frac{b-a}{c-a}$$

$$z = \frac{x-b}{c-a}$$

Segments will have length

$1-w$ or $w+z$

The optimal case is

$$z = 1-2w$$

$$\frac{z}{1-w} = w$$

This gives

$$w = \frac{3-\sqrt{5}}{2} = 0.38197$$


The number 0.38196 is related to the golden mean studied by Pythagoras

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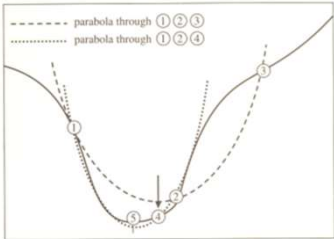
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Bracketing Interpolation

- Choose new point by interpolating the minima of a parabola.



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Bracketing Polynomial interpolation

- Bracket the minimum.
- Fit a quadratic or cubic polynomial which interpolates $f(x)$ at some points in the interval.
- Jump to the (easily obtained) minimum of the polynomial.
- Throw away the worst point and repeat the process.



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Optimization

- Univariate methods
- Multivariate methods
- Simulated annealing
- Genetic algorithms

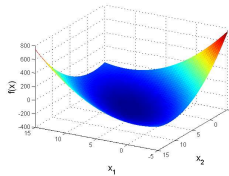
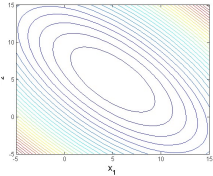
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Multivariate optimization

- Optimization of a function of multiple (N) parameters.
- We will consider examples for N=2, so that cost function surfaces can be visualized.

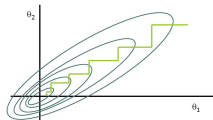
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Multivariate optimization

Single parameter enumeration

- Simple but inefficient approach
- Consider
 - Parameters $\theta = (\theta_1, \theta_2, \dots, \theta_k)$
 - Function $f(\theta)$
- Maximize θ with respect to each θ_i in turn
 - Cycle through parameters



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Multivariate optimization

Derivative based optimization

- Consider
 - Parameters $\theta = (\theta_1, \theta_2, \dots, \theta_k)$
 - Function $f(\theta; x)$
- Score vector

$$S = \frac{d \ln f}{d \theta} = \left(\frac{d \ln f}{d \theta_1}, \dots, \frac{d \ln f}{d \theta_k} \right)$$
 - Find maximum along $\theta + \delta S$

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Multivariate optimization

Steepest descent

- Basic principle is to minimize the N-dimensional function by a series of 1D line-minimizations: $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$
- The steepest descent method chooses $\mathbf{p}_k$ to be parallel to the gradient $\mathbf{p}_k = -\nabla f(\mathbf{x}_k)$



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Optimization

- Univariate methods
- Multivariate methods
- Simulated annealing
- Genetic algorithms



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Global minimum

Local Search algorithms

- Heuristic algorithms** – approach to problem solving that employs a practical method not guaranteed to be optimal or perfect, but sufficient for the immediate goals.
- Local search algorithms** – heuristic approach to optimization which starts from a candidate solution and then iteratively moves to a neighboring solution.
- Examples of local search algorithms:
 - Hill-climbing
 - Random walk
 - Simulated annealing



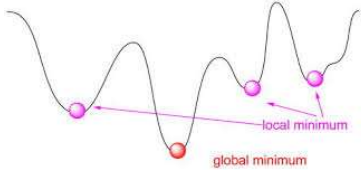
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Global minimum

- All the algorithms so far were useful for finding the local minimum.
- Simple approach: Test multiple initial points.



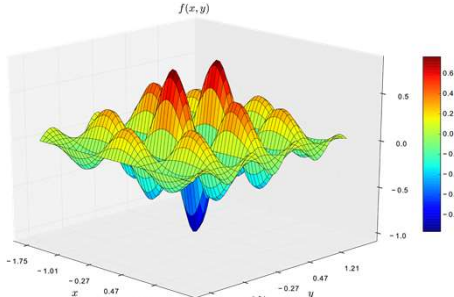


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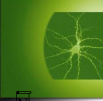

Multivariate problem is harder





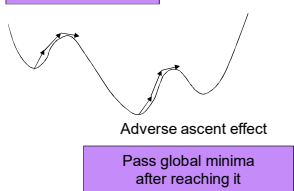
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Global minimum

Consequences of ascents





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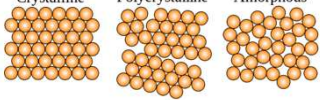

Annealing

- Annealing is a thermal process for obtaining low energy states of a solid in a heat bath.
- The process contains two steps:
 - Increase the temperature of the heat bath to a maximum value at which the solid melts.
 - Decrease slowly the temperature of the heat bath until the particles arrange themselves in the minimum energy state of the solid.

Crystalline

Polycrystalline

Amorphous










Simulated Annealing

- Start with a high temperature and a random state.
- Move to neighboring states:
 - Always accept moves that decrease the energy.
 - Only accept "bad" moves accordingly to a probability distribution dependent on the temperature.
- Decrease the temperature slowly, accepting less bad moves at each temperature level. (Very low temperatures → hill-climbing algorithm)
- The distribution used to decide if we accept a bad movement is known as Boltzman distribution.

$$P(\gamma) = \frac{e^{-E_\gamma/T}}{Z(T)},$$

$$Z(T) = \sum_{\gamma'} e^{-E_{\gamma'}/T},$$

γ is the current configuration of the system, E_γ is the energy related with it, Z is a normalization constant.

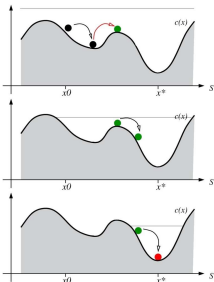







Simulated annealing

- Stochastic method
- Sometimes take up-hill steps to avoid local minima
- Solution is gradually frozen





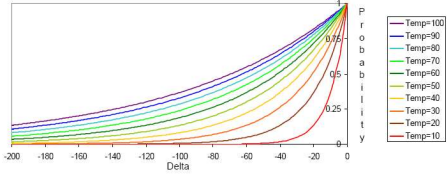





Simulated Annealing

- Acceptance criterion and cooling schedule

if (delta>=0) accept
 else if (random < $e^{\Delta E / T_{\text{curr}}}$) accept, else reject /* 0<=random<=1 */



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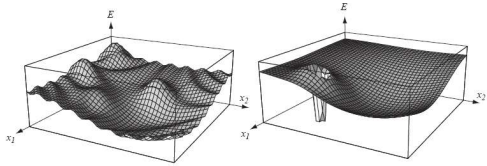





Simulated annealing

Practical Issues

- Cost function must be carefully developed, it has to be "fractal and smooth".
- The energy function of the left would work with SA while the one of the right would fail.



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Simulated annealing

Setting the parameters

- Initial temperature
 - Typically 50% of bad moves are accepted.
- Cooling steps
 - Reduces the temperature by 10%
- Number of iterations at each temperature
 - Move 1-10 times each "element" of the state.
- Final temperature
 - Typically should not accept any bad moves; this step is known as the **quenching** step.

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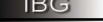
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Optimization

- Univariate methods
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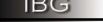





Components of a GA

A problem to solve, and ...

- Encoding technique *(gene, chromosome)*
- Initialization procedure *(creation)*
- Evaluation function *(environment)*
- Selection of parents *(reproduction)*
- Genetic operators *(mutation, recombination)*
- Parameter settings *(practice and art)*








Simple Genetic Algorithm

```

{
  initialize population;
  evaluate population;
  while TerminationCriteriaNotSatisfied
  {
    select parents for reproduction;
    perform recombination and mutation;
    evaluate population;
  }
}

```



